Week 12 – Lab Assignment

CSC 3320 – System-level Programming
Thursday April 9th, 2026

Introduction

Welcome to the 12th programming lab of CSC 3320! Today, we will be covering the following topics:

1. Structs
2. Separate Files for Structs

Lab Policies

- Attendance is mandatory.
- Labs must be completed **individually.**
- TAs are here to help you. Ask them for help!
- Lab assignments are due at midnight on the day of your lab.

Comments

The lab assignment requires the inclusion of comments to enhance code readability and understanding. Specifically, a block comment at the beginning of the C file is required. Your block comment should include the following:

- The program name
- The author's name (your name)
- A description of the program's overall purpose

Deliverables:

3. **VendingMachine.h** — Header file containing the struct definition and related function declarations.
4. **VendingMachine.c** — Source file containing the function definitions.
5. **main.c** — Source file containing the main() function.
6. A screenshot of the output in the Terminal.

If you have any questions, please do not hesitate to ask your TA.

Program: Vending machine

Given three integers as user inputs that represent the number of bottles of drinks to purchase, to restock, and to purchase afterward, create a VendingMachine variable that performs the following operations:

- Purchase the first input number of bottles of drinks
- Restock the second input number of bottles of drinks
- Purchase the last input number of bottles of drinks
- Report inventory

VendingMachine.h contains the struct definition and related function declarations. VendingMachine.c contains related function definitions. A VendingMachine's initial inventory is bottles of 20 drinks.

Ex: If the input is:

```
5 2 7
```

the output is:

```
Inventory: 10 bottle
```

File Structure:

You must separate your code into three files:

- **VendingMachine.h** — Contains the struct definition and related function declarations.
- **VendingMachine.c** — Contains the function definitions for all VendingMachine operations.
- **main.c** — Contains the main() function, reads user input, and calls the VendingMachine functions.

Deliverables:

For today's lab, you must upload all three C files and a screenshot of your output on iCollege. Please name your files as follows:

- **C Files**
  - `lastname_firstname_VendingMachine.h`
  - `lastname_firstname_VendingMachine.c`
  - `lastname_firstname_main.c`
  - For example: **`turing_alan_VendingMachine.h`**

- **Screenshots**
  - `lastname_firstname_vending_machine.png`
  - For example: **`turing_alan_vending_machine.png`**